Introduction to Sequences

Exam code: 0607

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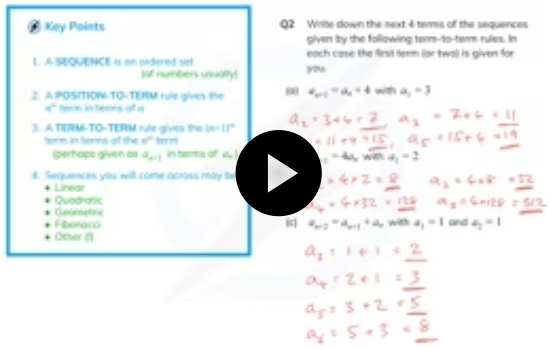

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Introduction to sequences

What are sequences?

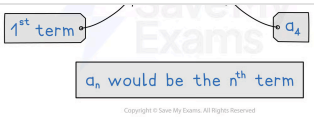
- A **sequence** is an ordered set of numbers that follow a **rule**
  - For example 3, 6, 9, 12...
    - The rule is to add 3 each time
- Each number in a sequence is called a **term**
- The **location** of a term within a sequence is called its **position**
  - The letter ***n*** is used for position
    - ***n*** **= 1** refers to the **1st term**
    - ***n*** **= 2** refers to the **2nd term**
    - If you do not know its position, you can say the ***n*** **th term**
- Another way to show the position of a term is using **subscripts**
  - A general sequence is given by $a_1, a_2, a_3, \ldots$
    - a_1 represents the **1st term**
    - a_2 represents the **2nd term**
    - a_n represents the ***n*****th term**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



How do I write out a sequence using a term-to-term rule?

- **Term-to-term rules** tell you how to get the **next term** from the term you are on
  - It is what you do **each time**
  - For example, starting on 4, add 10 each time
    - 4, 14, 24, 34, ...

How do I write out a sequence using a position-to-term rule?

- A **position-to-term rule** is an **algebraic expression** in ***n*** that lets you find **any term** in the sequence
  - This is also called the ***n* th term formula**

- You need to know what **position** in the sequence you are looking for
  - To get the **1st term**, substitute in $n = 1$
  - To get the **2nd term**, substitute in $n = 2$

- You can jump straight to the 100th term by substituting in $n = 100$
  - You do not need to find all 99 previous terms

- For example, the n th term is $8n + 2$
  - The 1st term is $8 \times 1 + 2 = 10$
  - The 2nd term is $8 \times 2 + 2 = 18$
  - The 100th term is $8 \times 100 + 2 = 802$

How do I know if a value belongs to a sequence?

- If you know the n th term formula, set the value equal to the formula
  - This creates an **equation** to **solve** for n

- For example, a sequence has the n th term formula $8n + 2$
  - Is 98 in the sequence?

$$8n + 2 = 98$$
$$8n = 96$$
$$n = \frac{96}{8}$$
$$n = 12$$

    - It is in the sequence, it is the 12th term

  - Is 124 in the sequence?

$$8n + 2 = 124$$
$$8n = 122$$
$$n = \frac{122}{8}$$
$$n = 15.25$$

Examiner Tips and Tricks

In the exam, it helps to write the position number (the value of n) above each term in the sequence.

Worked Example

A sequence has the n th term formula $3n + 2$.

(a) Find the first three terms in the sequence.

Answer:

Substitute $n = 1$, $n = 2$ and $n = 3$ into the formula

$$3 \times 1 + 2 = 5$$

$$3 \times 2 + 2 = 8$$

$$3 \times 3 + 2 = 11$$

5, 8, 11

(b) Find the 80th term.

Answer:

Substitute $n = 80$ into the formula

$$3 \times 80 + 2$$

The 80th term is 242

(c) Determine whether the number 96 is in the sequence.

Answer:

Set the formula equal to 96

$$3n + 2 = 96$$

Solve to find n
If n is a whole number, it is a term in the sequence

$$3n = 96 - 2$$
$$3n = 94$$
$$n = \frac{94}{3}$$

94 is not divisible by 3
The nearest possible value is 95 ((95−2) ÷ 3 = 31, the 31st term)

96 is not in the sequence

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